# Efficient Model Compression Techniques for Deploying Deep Neural Networks on Resource-Constrained Edge Devices

> A comprehensive study of model compression methods — pruning, quantization, knowledge distillation, and low-rank factorization — aimed at enabling efficient DNN deployment on edge devices.

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## Abstract

The rapid advancement of deep learning has enabled breakthroughs in computer vision, NLP, and speech recognition. However, deploying deep neural networks (DNNs) on resource-constrained edge devices remains a major challenge due to high computational complexity, memory requirements, and energy consumption.

This paper presents a comprehensive study of efficient model compression techniques and proposes an **integrated compression pipeline** that combines pruning, quantization, knowledge distillation, and low-rank factorization to achieve optimal trade-offs between model size, accuracy, latency, and energy efficiency.

**Keywords:** Model Compression · Edge Computing · Deep Neural Networks · Quantization · Pruning · Knowledge Distillation

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## Motivation

State-of-the-art deep learning models are designed for high-performance cloud environments where resources are abundant. Edge devices — smartphones, IoT devices, and embedded systems — operate under strict constraints:

- Limited processing power
- Restricted memory capacity
- Finite battery life

Model compression bridges this gap by reducing model size and complexity to enable efficient inference at the edge, without significantly compromising performance.

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## Compression Techniques

### Pruning

Removes unnecessary or less important weights from the neural network.

| Type | Description |
|---|---|
| Unstructured Pruning | Removes individual weights (introduces sparsity) |
| Structured Pruning | Removes entire neurons or filters (hardware-friendly) |

> **Note:** Unstructured pruning alone does not significantly reduce model size unless combined with quantization or compression-aware representations.

### Quantization

Reduces parameter precision from 32-bit floating point to lower bit-widths (e.g., 8-bit integers).

- Reduced memory usage
- Faster computation
- Lower energy consumption
- Dynamic quantization achieves ~**70% memory reduction** with negligible accuracy loss

### Knowledge Distillation

Trains a smaller **student** model to mimic the behavior of a larger **teacher** model using soft outputs, enabling comparable performance with fewer parameters.

### Low-Rank Factorization

Decomposes large weight matrices into smaller matrices, reducing computational complexity and storage — particularly effective for convolutional and fully connected layers.

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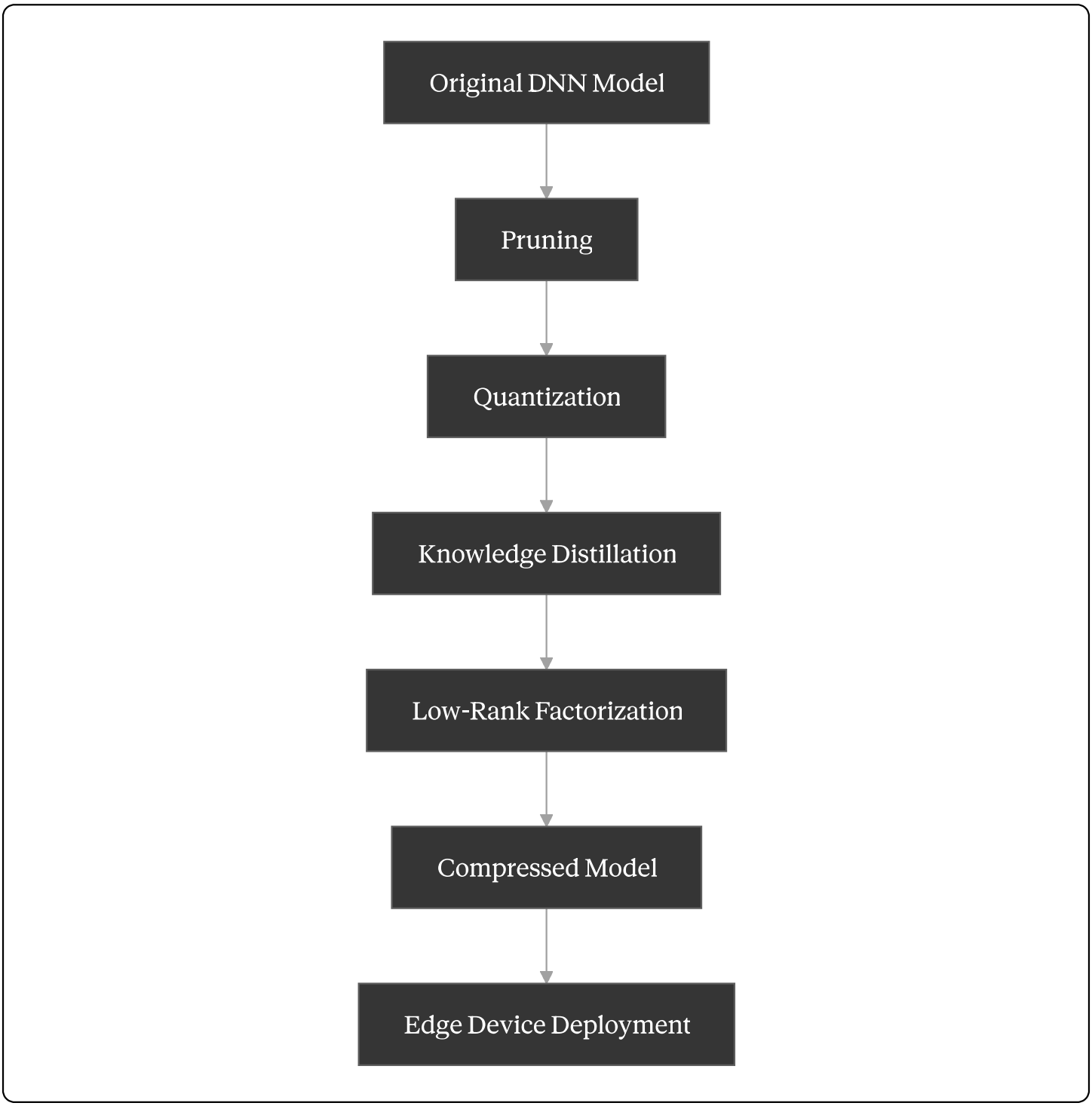
## System Architecture

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## Results

### Performance Gains (Hybrid Pipeline)

| Metric | Improvement |
|---|---|
| Model Size Reduction | **60–80%** |
| Inference Latency Reduction | **30–50%** |
| Memory Usage (Dynamic Quantization) | **~70% reduction** |

### Technique Comparison

| Technique | Size Reduction | Accuracy Loss | Latency Gain |
|---|---|---|---|
| Pruning | 40–60% | Low | Medium |
| Quantization | 50–70% | Medium | High |
| Distillation | 30–50% | Very Low | Medium |
| **Hybrid Approach** | **70–80%** | **Very Low** | **High** |

### Key Trade-off Insights

| Compression Setting | Outcome |
|---|---|
| Mild pruning (20–40%) | Maintains accuracy, reduces size |
| 8-bit quantization | Good balance of efficiency and precision |
| Distillation-enhanced models | Recovers most accuracy lost from other methods |

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## Experimental Setup

| Component | Details |
|---|---|
| **Datasets** | CIFAR-10, ImageNet subsets, text classification benchmarks |
| **Architectures** | CNNs, Transformer-based models |

| Component | Details |
| --- | --- |
| **Workloads** | Lightweight (simple classification), Moderate (image recognition), Heavy (deep CNN / long NLP) |
| **Compression Parameters** | Various pruning ratios, quantization bit-widths, distillation configs |

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## Future Scope

- **Adaptive compression** — dynamically adjusting model complexity based on available resources (memory, power, processing)
- **Federated learning integration** — privacy-preserving AI at the edge
- **Automated compression pipelines** — neural architecture search (NAS) and multi-objective optimization
- **Hardware-aware techniques** — tailoring compression strategies to specific edge hardware

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## References

Key papers this work builds on:


- Han et al., "Deep Compression," ICLR 2016
- Hinton et al., "Distilling the Knowledge in a Neural Network," arXiv 2015
- Jacob et al., "Quantization and Training for Efficient Inference," CVPR 2018
- Li et al., "Pruning Filters," ICLR 2017
- Satyanarayanan, "The Emergence of Edge Computing," Computer 2017
- Howard et al., "MobileNets," arXiv 2017


Full reference list available in the paper.

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